Genetic Endowments and the Long-Run Health Returns to Compulsory Schooling

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Abstract

Genetic differences tied to educational attainment predict later-life success and wellbeing. I study whether they also moderate how education affects physical health in old age. Using data on individuals aged 65 to 85 from the English Longitudinal Study of Ageing, I exploit the 1947 raising of the school leaving age in the UK from 14 to 15 in a fuzzy regression discontinuity setting to estimate the causal effect of schooling on frailty, the share of age-related health deficits. I estimate interactions with an educational attainment polygenic index (EA PGI), a DNA-based measure of genetic propensity for education. Health returns to schooling rise monotonically along the EA PGI, from zero in the lowest quintile to a 6 percentage point reduction in frailty in the highest. This gradient points to a complementarity between genetic endowments and education. These health improvements originate at the upper part of the frailty distribution. Suggestive evidence on channels indicates reduced smoking, higher physical activity, and a lower probability of receiving a disability pension.

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1 Introduction

Genetic differences between individuals are important drivers of success and wellbeing (Benjamin et al. 2012; Biroli et al. 2026). Their influence is already apparent in school, where achievement gaps by genetic propensity for education are as large as socioeconomic gaps (Houmark et al. 2025). Children with more of these genetic markers develop stronger cognitive and noncognitive skills (Belsky et al. 2016; Houmark, Ronda, and Rosholm 2024). Their influence also extends into adulthood and old age. Genetic propensity for education is correlated with occupation, income, and wealth (Papageorge and Thom 2020; Carvalho 2025), personality (Mõttus et al. 2017), health behaviors such as smoking (Wedow et al. 2018), and several severe diseases (Okbay et al. 2022).

Genetic propensity for education also affects later-life outcomes by moderating the effects of environments tied to education and skill accumulation. Children with at the higher end of the distribution acquire and retain skills more effectively, so that early advantages accumulate (Houmark, Ronda, and Rosholm 2024). They likely face lower marginal costs of schooling (Carvalho 2025) and therefore obtain more education (Hollenbach, Schmitz, and Westphal 2026). Furthermore, parents invest more in children with higher genetic propensity for education, reinforcing initial differences (Houmark, Ronda, and Rosholm 2024). Additionally, it affects the returns to different kinds of investments. For example, an additional year of schooling raises income and old-age cognitive abilities mainly for individuals with high propensity for education (Barcellos, Carvalho, and Turley 2021; Hollenbach, Schmitz, and Westphal 2026).

In this paper, I study how the effect of education on physical health in old age varies with genetic propensity for educational attainment. Using data on individuals aged 65–85 from the English Longitudinal Study of Ageing, I exploit the 1947 raising of the UK school leaving age from 14 to 15 in a fuzzy regression discontinuity design to estimate the causal effect of schooling. I measure health with a frailty index, the share of age-related health deficits present in an individual. To quantify genetic propensity, I use an educational attainment polygenic index (EA PGI), a DNA-based measure that sums the genetic variants an individual carries, each weighted by how strongly it is associated with schooling. It predicts a sizable share of the variation in educational attainment between individuals, but is not deterministic.

This paper contributes to the literature in two ways. First, I investigate the role of the EA PGI in health at old-age. To my knowledge, I am the first one to study the effect of interactions of educational attainment and genetic propensity for educational attainment in old-age health. Related studies on differential effects of education on later-life outcomes by genetic predisposition find that genes and education often reinforce each other. For health, the gains from education are largest among individuals at high genetic *risk* for the specific condition, including Alzheimer's disease and mid-life overweight (Barcellos et al. 2025; Barcellos, Carvalho, and Turley 2018), while the effects on metabolic health in middle age do not differ by the EA PGI (Barcellos, Carvalho, and Turley 2018). In contrast, the effects of education on income and on cognitive ability in old age increase in the EA PGI (Barcellos, Carvalho, and Turley 2021; Hollenbach, Schmitz, and Westphal 2026). I add to this by showing that

an early-life interaction of an exogenous increase in schooling and genetic propensity contributes to differences in health late in life.

Second, I contribute to the question of whether the typically large descriptive health differences by education reflect a causal effect, and if so, for whom and in which contexts. Evidence from UK compulsory schooling reforms is mixed. Clark and Royer (2013) estimate very small effects of the 1947 and 1972 reforms on self-rated health, and Jürges, Kruk, and Reinhold (2013) find no effects on self-rated health or biomarkers for inflammation. For the 1972 reform, Gehrsitz and Williams (2024) report no effect on subjective health but fewer hospitalizations among men, and Janke et al. (2020) document a lower probability of diabetes. Silles (2009) shows that education significantly increases self-reports of good health and of being free of (limiting) illness. Studies using UK Biobank data on the 1972 reform find that the additional education reduced body size at the upper end and increased blood pressure at the lower end of the respective distributions (Barcellos, Carvalho, and Turley 2023), and lowered the risks of diabetes and mortality (Davies et al. 2018). Except for Jürges, Kruk, and Reinhold (2013), these studies use cohorts that were still relatively young at the time, and all of them examine specific health measures, often chronic diseases or self-rated health. I instead study individuals up to age 85 and use the frailty index, which combines up to 56 diagnoses, conditions, limitations, and tests into a comprehensive measure of mostly functional health in old age.

My results first establish a sizable genetic health gap in old age. The difference in frailty between the highest and lowest quintile of the EA PGI is about as large as the difference between the first and the fourth quintile of socioeconomic status measured using an index including educational attainment, pension income and household wealth. Furthermore, frailty is lower among the more educated in every quintile of the EA PGI, but the health advantage of more schooling is concentrated with those in the highest PGI quintile. My fuzzy regression discontinuity estimates suggest that this pattern is also reflected in the causal effect of education. I find that health returns to an additional year of schooling rise monotonically along the EA PGI, from essentially zero in the lowest quintile to a 6 percentage point reduction in frailty in the highest. This gradient points to a complementarity between genetic endowments and education in old-age health. The gains come from the upper part of the frailty distribution. Additional schooling lowers the probability of being frail and I find suggestive evidence that for those in the highest PGI quintile, it also increases the probability of remaining nearly deficit-free for longer. Additionally, I find no evidence of selection on gains into schooling with respect to old-age frailty. Investigating potential channels, I show that the additional education from the reform improved smoking behavior and physical activity in old age and reduced the probability of receiving a disability pension. I do not find any impacts on old-age pensions, income or employment history.

The 1947 compulsory schooling reform increased educational attainment most among students with a low EA PGI, probably one of the reform's intended target groups. However, the additional education improved later-life health only for those with higher PGI scores. This suggests that this compulsory schooling reform may have been an ineffective tool for improving population health at old age. Furthermore, it is another piece of evidence that, while compulsory schooling may be able to compensate for

certain genetic risk factors (Barcellos, Carvalho, and Turley 2018), it exacerbates existing differences in the EA PGI, which has been shown to be a relevant source of inequality in school (Houmark et al. 2025; Biroli et al. 2026). Nevertheless, this does not mean that policy changes at other educational margins cannot support low-PGI students. One such margin could be school quality, which the 1947 reform did not improve (Clark 2023). Improvements in school quality may benefit later-life outcomes low-PGI students, since higher-quality schools have been shown to improve their academic achievement the most (Arold, Hufe, and Stoeckli 2025).

The remainder of this paper is structured as follows. Section 2 introduces the institutional setting of the 1947 UK compulsory schooling reform and the measure of genetic propensity for educational attainment. Section 3 describes the data and sample, the variables, and the empirical methods and assumptions. Section 4 presents the main results. I first show descriptive patterns, then estimate the interaction causally, and then investigate patterns of selection on gains. I close the section with robustness checks and suggestive evidence on non-health channels. Section 4.8 concludes.

2 Institutional setting and background

2.1 The 1947 raising of the school leaving age

To identify the causal effect of schooling on late-life health, I exploit the 1947 compulsory schooling reform in the UK. A second reform raised the leaving age from 15 to 16 in September 1972. I do not use this reform because cohorts affected by it are still too young in the data to study late-life health outcomes. Furthermore, genetic data have only been collected once in the English Longitudinal Study of Ageing, covering mostly the older cohorts.

The 1944 Education Act raised the minimum school-leaving age from 14 to 15, with implementation set by Ministerial order to April 1, 1947 (Clark and Royer 2013). Because children in the UK typically started school at age five, the reform extended the mandatory length of schooling from nine to ten years. Cohorts born from April 1933 onward, the pivotal cohort, were the first bound by the new rule. I later on exploit this discontinuity by comparing cohorts born before and after the cutoff.

The reform had a strong and broad effect on educational attainment, shifting a large share of the students from nine to ten years of schooling. Figure 1 illustrates this shift using cohort-level aggregates from ELSA. The figure plots the share of individuals still enrolled at ages 15 and 16 by birth cohort, with a vertical line marking the pivotal cohort. The top line (circles) tracks enrollment at age 15. For unaffected cohorts, this share sits around 40%, implying that roughly 60% of students left at the previous compulsory leaving age of 14. Both series trend upward in the cohorts just before the cutoff, but only enrollment at 15 jumps sharply at the pivotal cohort, rising from around 60% to nearly 100%. Enrollment at age 16 (diamonds) shows no such discontinuity. Compliance was high, but not complete. Some individuals report leaving school at age 14 even after the reform took effect. As Clark and Royer (2013) documents, this noncompliance is concentrated among individuals born in the summer months, who turned 15 between the end of one school year and the start of the next, and could therefore leave at age 14.

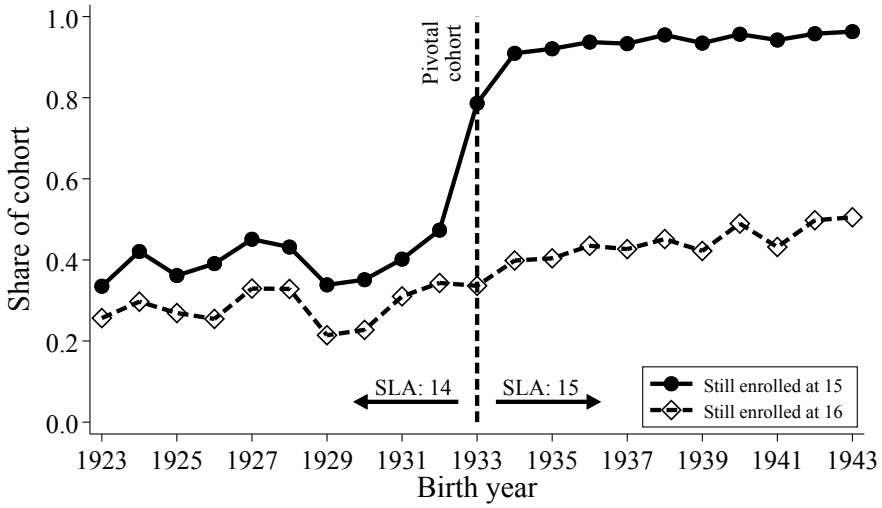


FIGURE 1 – School enrollment at ages 15 and 16 by birth cohort

Note: The 1944 Education Act raised the minimum school leaving age (SLA) from 14 to 15 for cohorts born from April 1933 onward. This figure shows yearly cohort means for enrollment at 15 and 16 since information on month of birth are unavailable. The mean of the pivotal cohort of 1933 therefore includes both individuals affected and unaffected by the schooling reform. The share still enrolled at age 16 is shown for comparison. Sample: 10,709 individuals across waves 1-10 of the English Longitudinal Study of Ageing.

The reform raised the quantity of schooling without changing its quality. Nearly all compliers attended lower-track schools, which emphasized practical education and were of lower quality than grammar schools in terms of class size and teacher qualifications (Clark 2023). School resources adjusted to accommodate the increased enrollment, so these inputs did not change as a result of the reform (Clark and Royer 2013). The additional year did not raise the probability of obtaining formal academic or vocational qualifications on average (Clark 2023). Instead, schools used the extra year to extend the existing curriculum, introducing some students to more advanced material and helping others consolidate basic skills (Clark and Royer 2013). The effect of the reform can therefore be thought of as an increase in quantity of schooling.

The 1947 reform has been used to identify causal effects of education on wages (Harmon and Walker 1995; Oreopoulos 2006; Devereux and Hart 2010; Clark and Royer 2013), other labor market outcomes (Clark 2023), financial behavior (Gray, Montagnoli, and Moro 2021), health (Silles 2009; Powdthavee 2010; Clark and Royer 2013; Jürges, Kruk, and Reinhold 2013), intergenerational health (Lindeboom, Llena-Nozal, and van der Klaauw 2009; Silles 2011), health knowledge (Johnston et al. 2015), mortality (Clark and Royer 2013; Gathmann, Jürges, and Reinhold 2015), and cognitive ability (Banks and Mazzonna 2012; Hollenbach, Schmitz, and Westphal 2026).

2.2 Measuring genetic propensity for education

Frequently used measures of student ability use final products of that ability, such as IQ tests results, scores on academic or vocational tests, and college admission decisions (see, e.g., Brinch and Galloway 2011; Neal and Johnson 1996; Black and Smith 2006; Dale and Krueger 2002). Perhaps more cautiously

interpreted as a propensity for education or learning, these measures also have in common that they are malleable outputs of complex inputs early in life, such as genes, conditions and investments (Todd and Wolpin 2003; Cunha and Heckman 2007).

In this paper, I rely on a DNA-based measure that has the advantage of being fixed at conception and not modifiable afterwards. Its use for complex traits like educational attainment is grounded in two basic facts about human genetics. First, humans share most of their genome, but small differences can explain variation in traits. The human genome consists of roughly three billion DNA base pairs, and any two individuals share more than 99 percent of their DNA on average (Auton et al. 2015; Nurk et al. 2022). The remaining differences account for a substantial share of the variation in observable traits and behavior: a meta-analysis of twin studies by Polderman et al. (2015) finds that genetic factors explain about half of the population variation across 2,929 traits. Second, There is no single gene for a complex trait. Genetic influence on traits and behavior is mostly polygenic, meaning it is spread across many locations in the genome that each contribute a small effect (Chabris et al. 2015; Visscher et al. 2017). Education fits this pattern, with thousands of genetic variants individually associated with years of schooling, each contributing only a small share (Lee et al. 2018; Okbay et al. 2022).

The standard tool for summarizing small additive contributions of many genetic variants into one measure is the polygenic index (PGI). Most genetic differences between individuals come down to single positions in the genome where the DNA base varies across the population. These are called single-nucleotide polymorphisms (SNPs). A PGI summarizes an individual's genetic predisposition for a given outcome by summing their SNPs, each weighted by its estimated association with that outcome. To quantify SNPs, one variant present in the population is chosen as the reference variant. Any person i can inherit up to two copies of each SNP j, one from each parent and therefore carries either zero, one, or two copies of the reference variant $r_{ij} \in \{0, 1, 2\}$. Weights w_j, i.e. the association of each SNP with the outcome, are obtained in large discovery studies, called genome-wide association studies (GWAS). They estimate how strongly each SNP is correlated with the trait of interest by regressing the trait on each SNP, one at a time, in a large separate sample to avoid overfitting.

The PGI is then simply the weighted sum of counts of the reference variant across all SNPs:

$$PGI_i = \sum_j w_j r_{ij} \tag{1}$$

The PGI is a single individual-level index of additive genetic probability for a certain outcome. In this paper, I use a PGI for educational attainment (EA PGI). It is approximately normally distributed, with individuals at the upper end carrying more of the alleles that, on a population level, are positively associated with schooling and those at the lower end carrying fewer. Even though, at the individual level, it is informative about educational attainment, the index is not deterministic, because its realization depends on early-life environments. Individuals with a high EA PGI can have low attainment, and vice versa. Therefore, the EA PGI can not directly be interpreted as innate ability. Furthermore, it captures more than ability, including traits like motivation or part of family environments (Kong et al.

2018; Demange et al. 2021). Instead, it can be understood as an index of genetic factors relevant for human capital accumulation (Barth, Papageorge, and Thom 2020).

3 Data and methods

3.1 Data and variables

I use data from the the English Longitudinal Study of Ageing (ELSA), a nationally representative longitudinal survey of adults aged 50 and older living in England (Banks et al. 2026). ELSA has been conducted biennially since 2002 and includes detailed information on physical and mental health, socioeconomic status, and, over time, has integrated biological and genetic data. For more information, see Steptoe et al. (2013).

Health in old age. My outcome variable is the frailty index, the share of potential age-related health deficits that an individual exhibits in a given survey wave. Deficits are symptoms, signs, diagnosed diseases, disabilities, and functional impairments. The index therefore summarizes health across many dimensions in a single number bounded between zero and one. It is grounded in the view of aging as an accumulation of deficits and a gradual loss of the body's redundancies (Mitnitski, Mogilner, and Rockwood 2001; Dalgaard and Strulik 2014). A higher value indicates a more advanced state of aging rather than the presence of one particular condition. A key characteristic of the frailty index is its ability to capture health status and its variation with high granularity. Compared with binary or ordinal health measures common in applied work, it is continuous and finely graded, and it relies on many concrete items instead of a single overall self-assessment (Hosseini, Kopecky, and Zhao 2022).

The frailty index is an established measure of health. First introduced in gerontology (Mitnitski, Mogilner, and Rockwood 2001) and standardized by Searle et al. (2008), it predicts mortality, institutionalization, and further adverse health events, and it predicts these outcomes more accurately than self-reported health status (Rockwood and Mitnitski 2007; Hosseini, Kopecky, and Zhao 2022). Results also do not depend on the exact items chosen. As long as enough deficits from different areas of health enter, indices built from different item lists behave similarly (Searle et al. 2008; Hosseini, Kopecky, and Zhao 2022). Economists have taken up the measure both as the health state in models of aging (Dalgaard and Strulik 2014; Strulik 2018b) and as the health outcome in empirical work, in studies of aging patterns across Europe (Abeliansky and Strulik 2018), of retirement and occupational health gaps (Abeliansky and Strulik 2023), and of the role of health in life-cycle earnings inequality (Hosseini, Kopecky, and Zhao 2022, 2025).

I follow the standard procedure of Searle et al. (2008), implemented in ELSA, for example by Marshall et al. (2015) and Niederstrasser, Rogers, and Bandelow (2019), and use a broad set of 56 deficits available in ELSA. They include diagnoses of serious illnesses, difficulties with activities of daily living and mobility, sensory impairment, self-rated health, depressive symptoms, and results from cognitive tests. Each item takes a value between zero and one. Binary items, such as diagnoses, equal one if the deficit is present and zero otherwise. Ordinal items, such as self-rated health, can also take

intermediate values. The index is the sum of all 56 items divided by 56. For instance, an individual with 14 of the 56 deficits has a frailty index of 0.25. Appendix Table A.1 lists all items and how each is coded.

Education. ELSA does not record years of schooling directly. Instead, it reports the age at which a respondent left full-time education. My measure of educational attainment is an indicator variable equal to 0 if an they left school at age 14 or below and 1 if they stayed until at least age 15, i.e., left school at 15 or older. This is the margin targeted by the compulsory schooling reform. Most individuals in the final sample have low educational attainment, which was typical for these birth cohorts and one reason for the compulsory schooling reform (Oreopoulos 2006; Woodin, McCulloch, and Cowan 2013). Twenty-one percent left school at age 14 or earlier (the compulsory schooling age before the reform), 35 percent at age 15 (the compulsory schooling age after the reform), 30 percent at ages 16 to 18, and 13 percent at age 19 or later.

Schooling reform. The compulsory schooling reform took effect in April 1947 and applied to everyone born on or after 1 April 1933. Accordingly, I define the instrument that exogenously shifts educational attainment as an indicator variable equal to 0 if an individual was born before 1933 and 1 if they were born in 1933 or later. ELSA does not provide birth months, so for individuals born in 1933 it is unclear whether they were subject to the new school leaving age. Because almost all individuals are interviewed multiple times and in different months, I can combine birth year, interview month, and age at interview to infer for part of the 1933 cohort whether they were born between January and March (pre-reform) or between April and December (reform). I drop the remaining individuals born in 1933, whom I cannot definitively assign to either side of the reform.

Genetic propensity for educational attainment. To measure the genetic propensity for educational attainment, I use an EA PGI, as introduced in Section 2.2. The PGI is readily available in ELSA (NatCen Social Research 2026) and the weights used to calculate it are based on the discovery study by Lee et al. (2018) covering roughly 1.1 million individuals. Lee et al. (2018) report that it can explain 11–13 percent of the variation in educational attainment and about 10 percent of the variation in cognition in different samples.

Figure 2 shows the distribution of the standardized EA PGI in my final sample by the educational attainment indicator I use in this paper (left school at 14 or below versus 15 or later). While both groups have individuals with high and low EA PGI and most mass in the middle, the distribution skews towards lower values in the group leaving school earlier. To investigate interactions of schooling and the PGI at different parts of its distribution, I divide the score into quintiles.

Control variables. Since I am only interested in the differential effect of education on frailty by EA PGI, it would be sufficient to use the PGI as a pre-determined (and therefore pre-treatment) interaction variable. Nevertheless, I include parental education as additional control variables, in line with Barth, Papageorge, and Thom (2020), Papageorge and Thom (2020), and Barth et al. (2022), among others.

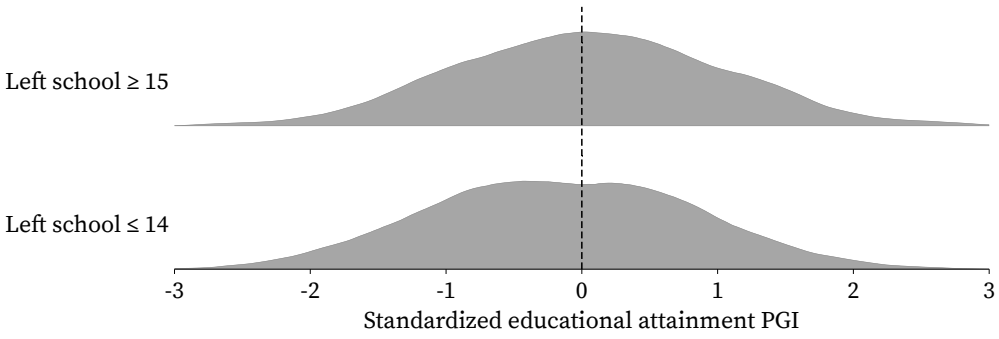


FIGURE 2 – Distribution of the EA PGI by treatment status

Note: Kernel density plots of the EA PGI distribution by educational attainment indicator (left school at 15 or later). Based on the estimation sample covering 3,415 individuals from the English Longitudinal Study of Ageing. Sample restrictions as described in Section 3.2

This is because the association between an individual's EA PGI and their schooling does not reflect a clean causal effect of the PGI. Part of the association is due to genetic nurture, whereby parents pass on both genes and a home environment. Parents whose own genes are associated with more schooling also tend to provide a home environment that supports learning. The child's PGI is therefore correlated with their schooling partly because it proxies for family environment, not only because of the child's own DNA (Kong et al. 2018). Conditional on the parents' genes, the child's genes are exogenous. Therefore, controlling for parental genotype would yield a causal effect of the PGI. A second strategy is a within-family design that compares siblings. Because they share the same parents, family fixed effects absorb parental genotype. Few datasets to date contain parent-child trios or enough siblings for a well-powered analysis. In ELSA, only between-family designs are feasible. In that case, Houmark, Ronda, and Rosholm (2024) show that controlling for parental education can absorb almost all of the family environment component.

I control for parental education by including indicator variables for the age at which each parent left full-time education. In ELSA, these ages are censored at 14 and 19. This information is missing for 17.5 percent (mothers) and 18.1 percent (fathers) of my final sample. Instead of dropping these observations, I include an additional category for missing parental school-leaving age.

Furthermore, I add controls for population stratification. Because of historical migration and non-random mating, subpopulations differ systematically in the frequency of genetic variants. If they also differ in schooling or health for cultural or environmental reasons, the discovery study underlying the PGI assigns weight to variants that signal ancestry rather than the trait itself. The PGI then partly captures the environments associated with ancestry, such as region, language, or local socioeconomic conditions, in a similar way as it partly captures the family environment. I therefore follow the convention in this literature (see e.g. Barth, Papageorge, and Thom 2020; Papageorge and Thom 2020; Barth et al. 2022) and additionally control for the first ten principal components of the genetic data. These are summary indices of general genetic similarity between individuals and account for population stratification (Price et al. 2006). ELSA provides them as pre-computed variables.

3.2 Sample and summary statistics

The analysis sample includes all ELSA respondents with genetic data who were born between 1923 and 1943, i.e. within ten years of the pivotal cohort of the compulsory schooling reform. They are observed at ages 65–85 in waves 1–10 of ELSA. From the 1933 cohort itself, I drop individuals whom I cannot assign to either side of the April 1 cutoff based on birth year, interview month and reported age at interview. These restrictions leave 18,705 observations from 3,415 individuals. Section 4.6 reports robustness checks that vary these sample restrictions.

Table 1 shows summary statistics for my sample, both in its entirety and separately by treatment and control group. The average frailty index is 0.16, with a substantial standard deviation. A majority of the sample consists of treated observations and individuals born after the reform cutoff. Treatment and control group differ significantly along several dimensions. Some of these differences are mechanical, most notably birth year and being born after the cutoff. Others concern individual characteristics: the treated group has a slightly higher share of women, is slightly younger, and is more concentrated in the upper part of the EA PGI distribution. The age gap is smaller than the gap in birth years, as repeated measurements at different ages seem to offset the latter to some degree. Crucially, individuals in the treated group are significantly less frail.

TABLE 1 – Summary statistics

	Full sample		By E_i (Left school at 15 or later)		
	Mean	SD	0	1	Diff. (p-val.)
Outcome					
Frailty index	0.16	0.11	0.19	0.15	−0.04 (0.00)***
Treatment					
Left school at 15 or later	0.79		0.00	1.00	1.00
Instrument					
Born 1933 or later	0.68		0.14	0.83	0.69 (0.00)***
Educational attainment PGI					
1st quintile	0.19		0.23	0.17	−0.06 (0.00)***
2nd quintile	0.20		0.21	0.19	−0.02 (0.01)***
3rd quintile	0.20		0.19	0.20	0.01 (0.05)**
4th quintile	0.21		0.21	0.21	0.00 (0.58)
5th quintile	0.21		0.16	0.22	0.06 (0.00)***
Selected controls					
Female	0.54		0.53	0.55	0.02 (0.01)***
Age	74.24	5.34	77.05	73.49	−3.56 (0.00)***
Birth year	1,935.03	5.24	1,929.80	1,936.42	6.62 (0.00)***
Observations	18,705		3,927	14,778	

Note: Means and standard deviations (continuous variables) for the full estimation sample, as well as means for the treatment ($E_i = 1$) and control ($E_i = 0$) groups. The last column reports the raw difference in means and p-values of a t-test for equality of means. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels. The full set of summary statistics, additionally including parental education, survey wave, and principal components of the genetic data, is reported in Appendix Table A.2.

3.3 Method

To estimate the causal effect of schooling on old-age frailty and how it varies with genetic propensity for education, I implement a fuzzy regression discontinuity design, estimated by two-stage least squares (2SLS). The second stage is

$$Y_{it} = \sum_{g=1}^{5} \Big(\alpha_g \, \mathbb{1}[G_i = g] + \beta_g \, E_i \times \mathbb{1}[G_i = g] \Big) + \theta_1 c_i + \theta_2 \, Z_i \times c_i + X_{it}' \gamma + \varepsilon_{it}. \tag{2}$$

The outcome Y_{it} is the frailty index of individual i in survey wave t. E_i indicates whether i stayed in school until at least age 15, and $G_i \in \{1, \ldots, 5\}$ denotes i's quintile of the EA PGI. The quintile indicators $\mathbb{1}[G_i = g]$ let average frailty differ across quintiles, and the interactions $E_i \times \mathbb{1}[G_i = g]$ let the effect of schooling vary across them.

Since schooling is endogenous, I instrument each interaction $E_i \times \mathbb{1}[G_i = g]$ with $Z_i \times \mathbb{1}[G_i = g]$, where Z_i indicates birth in 1933 or later, the first cohort affected by the 1947 increase in the minimum school-leaving age to 15. Instrumenting via 2SLS yields one first stage per PGI quintile:

$$E_i \times \mathbb{1}[G_i = g] = \alpha_g^{fs} \, \mathbb{1}[G_i = g] + \pi_g \, Z_i \times \mathbb{1}[G_i = g] + \theta_{g1}^{fs} c_i + \theta_{g2}^{fs} \, Z_i \times c_i + X_{it}' \gamma_g^{fs} + u_{git}, \quad g = 1, \ldots, 5. \tag{3}$$

The coefficient π_g is the jump at the 1933 cutoff in the share of individuals in quintile g who stayed in school until at least age 15, that is, the complier share in that quintile. The 2SLS estimation replaces each interaction $E_i \times \mathbb{1}[G_i = g]$ in Equation 2 with its predicted value from the first stage in Equation 3. Each β_g then identifies the local average treatment effect (LATE) of schooling on frailty for compliers in quintile g. Differences across the coefficients β_g capture how the effect of E_i on Y_{it} varies along the EA PGI distribution and thus reveal patterns of gene-environment interaction.

Both stages include the running variable c_i = birth year – 1933, a linear cohort trend centered at the 1933 cutoff, and its interaction with Z_i. The interaction allows the slope to differ on either side of the cutoff. X_{it} contains indicator variables for gender, age, survey wave, and mother's and father's school-leaving age, as well as the first ten principal components of the genetic data as controls. Standard errors are clustered at the individual level.

3.4 Assumptions and identification

Assumptions. Identification rests on three main assumptions. The first is that the 1947 reform is a valid instrument for education. Instrument relevance is not in doubt: the higher leaving age induced a large share of students to increase their attainment (see Figure 1), and my first stage is correspondingly strong (see Section 4.5). Instrument exogeneity is also likely to hold, given that eligibility for the compulsory schooling reform depends on date of birth and that the 1944 Education Act came years after these cohorts were born.

Second, I assume that the exclusion restriction holds. This requires that nothing but compulsory schooling shifts at the threshold. The leading concern is exposure to the Second World War. Both Clark

and Royer (2013) and Banks and Mazzonna (2012), who also use this reform, conclude that wartime disruptions did not vary systematically across the cutoff. The cohort trends I include absorb gradual variation in such exposure, and my estimates hold for narrower bandwidths (see Section 4.6). The exclusion restriction could also be violated if the reform affected always-takers, for example through spillovers if it changed not only instruction time but also school quality. However, the compliers in this particular reform were almost exclusively students in lower-track schools that ended at the minimum leaving age (Clark 2023). This arguably left little scope for spillovers onto those who would have stayed on regardless. Furthermore, as enrolment expanded, so did school resources, leaving class sizes and teacher qualifications largely unaffected (Clark and Royer 2013). The additional year of schooling did not yield formal qualifications, which would have required remaining in school until at least age 17 (Banks and Mazzonna 2012; Clark 2023). The reform lengthened schooling without changing its character.

The third assumption, monotonicity, rules out defiers, that is, individuals who left school earlier only because they were subject to an additional compulsory year. Defiance of this kind seems unlikely.

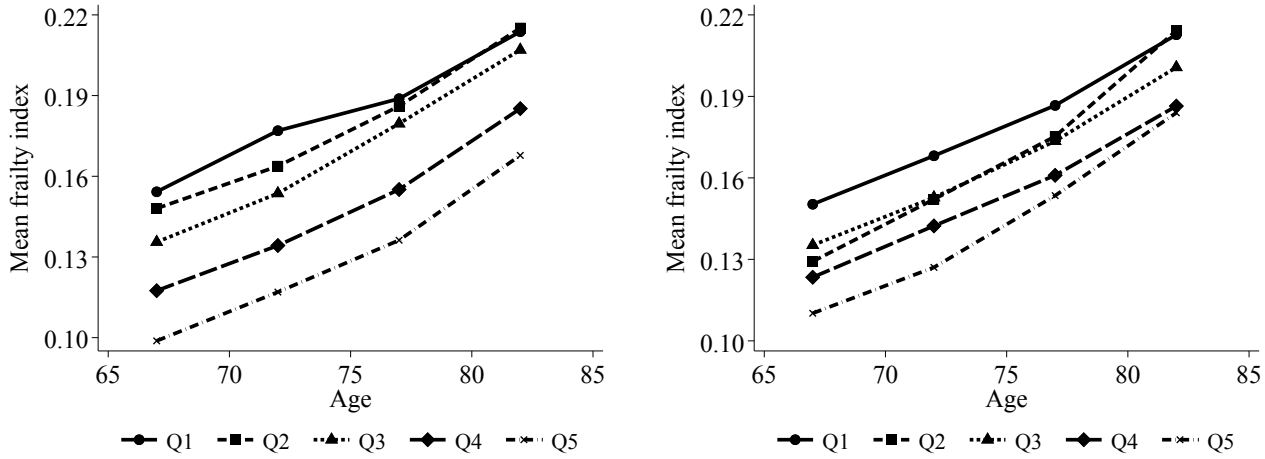
Attrition. In longitudinal studies of older individuals, such as ELSA, nonrandom attrition may be problematic if it is related to the treatment. If differential attrition is present, those who remain in the sample skew the treatment (or control) group, potentially biasing the estimated treatment effect. For this study, a While the debate on a causal effect of compulsory schooling on mortality is not settled and depends, among other things, on institutional setting (Galama, Lleras-Muney, and van Kippersluis 2018), Clark and Royer (2013) find no effect on mortality from the 1947 compulsory schooling reform. Even if differential survival were present, the direction of the resulting bias is predictable in my application. Since frailty increases strongly with age, longer survival in my treated group would mean that more frail individuals stay alive, shifting the treatment group towards higher frailty. Given that I find a reduction in frailty due to education, this means I would likely underestimate the true effect. I nevertheless test for panel attrition and find no evidence of it, and I estimate my main specification, reweighted to account for attrition. See Section 4.6 for details.

4 Results

4.1 Descriptive patterns

In this section, I first show that the EA PGI is generally informative about health differences later in life. I then document that the educational health gap in my setting varies with the EA PGI in a way that is already consistent with the gene-environment interaction in the causal effect of education on health that I estimate in the next section.

The EA PGI is designed to predict educational attainment, although, depending on the discovery study and cohort, it currently explains only about 11 to 16 percent of the variation in educational attainment (Lee et al. 2018; Okbay et al. 2022). Nevertheless, genetic achievement gaps in school are as large as socioeconomic achievement gaps and follow the same trajectory across grades (Houmark



A. Socioeconomic differences in old-age frailty

B. EA PGI differences in old-age frailty

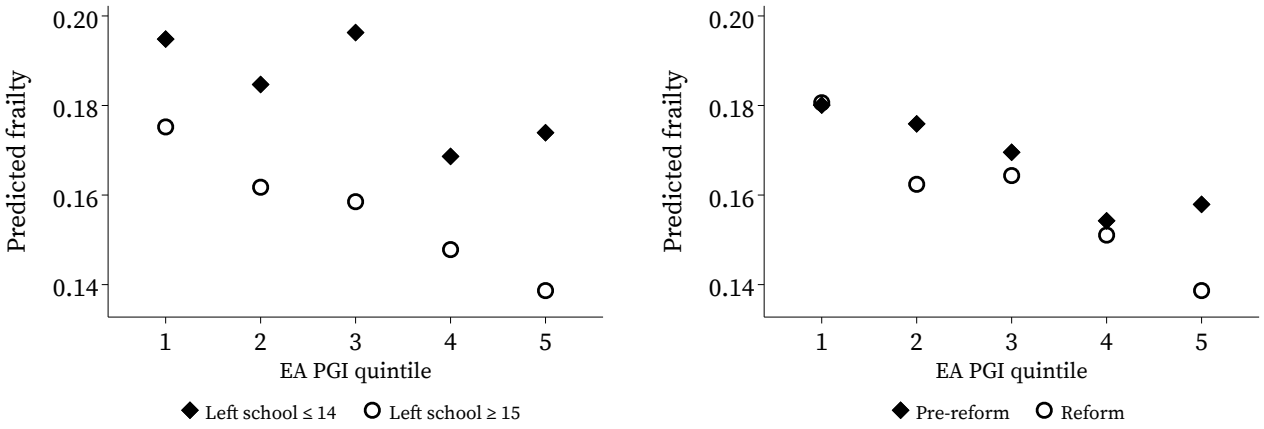
FIGURE 3 – Socioeconomic and genetic frailty gaps

Note: Progression of the average frailty index from age 65 to 84 in 5-year bins and separately by quintile of a socioeconomic score (Panel A) that includes education, household wealth and pension income and quintile of the EA PGI (Panel B). Averages are plotted at the midpoint of each 5-year bin. Sample: 26,850 observations from 6,083 individuals

et al. 2025). The EA PGI also appears to capture information relevant to outcomes beyond schooling. It has a causal effect on occupation, income, and home ownership (Carvalho 2025). It is correlated at the population level with personality (Mõttus et al. 2017), non-cognitive skills (Belsky et al. 2016), early-life physical and mental health (Hughes et al. 2024; Bolyard and Savelyev 2025), and health behaviors such as smoking (Wedow et al. 2018). It further predicts diagnoses of ten cardiometabolic, lung, musculoskeletal, neurological, and psychiatric diseases (Okbay et al. 2022). Taken together, these findings may point to pleiotropy, where the genetic variants in the EA PGI affect multiple outcomes, either directly or through their effect on educational attainment.

Given these propeties of the EA PGI, Figure 3 compares health gaps and trajectories from age 65 to 84 by socioeconomic status (SES) and EA PGI. It shows means of the frailty index for 5-year age bins, separately for quintiles of an SES index (Panel A) and quintiles of the EA PGI (Panel B). I construct the SES index following Houmark et al. (2025), but adapt it to older individuals. It uses the level of education, total household wealth and pension income. Household wealth is broken down by dividing by the square root of household size. For each of the three components, I rank respondents within each wave of data collection and convert the ranks to percentiles. Ranking within wave ensures that the index is unaffected by inflation or by respondents being observed in different waves. I then average each respondent's percentile rank over all waves and take the average of the three percentile ranks to get the SES index. Each component receives equal weight and by averaging, there is one observation per respondent, so that the cut-offs do not depend on how many times each person was observed. Finally, I assign each respondent to a quintile of the index.

Figure 3 makes two points. First, the EA PGI itself is informative about late-life health. As with SES, frailty differs clearly across EA PGI quintiles, with those in the highest quintile being the healthiest



A. Average frailty by educational attainment and EA PGI

B. Average frailty by reform exposure and EA PGI

FIGURE 4 – Predicted frailty by educational attainment, reform exposure and EA PGI quintile

Note: Panel A: Predicted frailty index by education indicator and quintile of the EA PGI. Panel B: Predicted frailty index by exposure to the 1947 UK compulsory schooling reform, the instrument. Predictions are non-parametrically adjusted for age and gender.Variable definitions according to section 3.1. Based on the main estimation sample as described in Section 3.2 with 18,705 observations from 3,415 individuals.

(lowest frailty) and those in the lowest quintile the frailest. These differences are already present at ages 65 to 69 and widen almost in parallel with age. Second, the frailty gap by EA PGI is almost as large as the SES gap. The total frailty difference by EA PGI (between quintiles 1 and 5) matches the frailty difference between SES quintiles 1 and 4.

Next, I investigate descriptive patterns of gene-education interactions by illustrating that the health differences between more and less educated individuals in my sample vary with the EA PGI. Figure 4 shows predicted frailty, adjusted for age and gender, across quintiles of the EA PGI. Panel A splits the sample by whether individuals left school at age 15 or later, and Panel B by exposure to the 1947 compulsory schooling reform. Panel A reveals three patterns. First, individuals with more schooling are healthier than those with less schooling. Second, frailty decreases in the EA PGI, as documented above. Third, this decrease is steeper for those with more schooling. In this group, individuals at the high end of the EA PGI are considerably less frail than those in quintiles 1–3. Panel B compares groups defined by exposure to the compulsory schooling reform. The same patterns hold, but they are less pronounced. Differences between the two groups are smaller in every PGI quintile. There is little to no difference between the two groups in quintiles 1, 3, and 4, but still a considerable difference in quintiles 2 and 5.

Overall, this hints at potential interactions between the EA PGI and education in late-life frailty: those with more education are less frail, but how much less differs by quintile of the EA PGI and those with more education and in the highest quintile are particularly healthy. These patterns are descriptive and do not yet imply a causal effect of (compulsory) schooling on frailty, nor that such an effect would vary by the EA PGI. Instead, it may be that generally healthier individuals are acquiring more education and that those who are healthier in early life are also healthier in later life. Since both educational

attainment and early-life health are correlated with the EA PGI, these descriptive patterns could reflect simple correlations and reverse causality.

4.2 The causal effect of education on frailty by EA PGI

To estimate a causal effect of education on frailty and investigate whether it varies by the EA PGI, I proceed to estimate a fuzzy regression discontinuity model, as introduced in Section 3.3. I present the results alongside an results from an OLS regression in Table 2. I show both estimation results for an average effect without interacting by EA PGI and an interacted regression that estimates the relationship between my education indicator and the frailty index in each quintile of the EA PGI according to Equation 2.

The OLS regressions in columns (1) and (2) show a statistically significant negative correlation between the treatment E_i (staying in school until at least age 15) and old-age frailty, both in the pooled regression and in each PGI quintile. Across all quintiles of the EA PGI, staying in school for the additional year is associated with a decrease in diagnosed conditions and impairments by about 2 – 4 percentage points. All estimates are relatively small, but they consistently have a negative sign.

The IV estimate in Column (3) of Table 2 indicates that, on average across the EA PGI, an additional year of education reduces frailty by about 3 percentage points, 18 percent of the mean. Column (4) shows that this average is mainly driven by individuals in the highest quintile of the EA PGI, whose frailty decreases by almost 6 percentage points (37 percent of the mean). This is the only statistically significant interaction effect, suggesting that more education reduces the share of diagnosed conditions only among those with a high EA PGI.

The paper's main contribution, however, is to document how the effect of education on old-age frailty changes along the EA PGI. The estimates in Column (4) increase in magnitude with the EA PGI. The effect is practically zero in the lowest quintile. The coefficients for quintiles 2 to 4 are larger but similar to each other, and they are about half the size of the coefficient for the highest quintile. Although these middle coefficients are not statistically significant, they are potentially economically relevant. On average, moving up one PGI quintile increases the effect of education on old-age frailty by 1.2 percentage points of the frailty index.[1] This pattern is consistent with a gene-education complementarity, where those who have more genetic variants strongly correlated with educational attainment and learning benefit more from increased instruction time.

Since a LATE corresponds to the ratio between reduced form and first stage, a large coefficient could in principle be driven by a small first stage. The first stage, reported in Table A.3 does indeed decrease in EA PGI, but remains large, even in the highest quintile, with about 36 percent being compliers. Furthermore, the reduced-form estimates in Column (5), where frailty is regressed directly on interactions between the instrument and EA PGI quintiles, suggest the same pattern as the IV results. There is a statistically significant effect of eligibility for the compulsory schooling reform on

[1] This is the average change between adjacent quintiles, $(\beta_5 - \beta_1)/4 = (-0.059 - (-0.010))/4 \approx -0.012$.

TABLE 2 – The effect of education on old-age frailty by EA PGI

DV: Frailty index	(1) OLS	(2) OLS	(3) IV	(4) IV	(5) RF
E_i	−0.028***		−0.029**		
	(0.005)		(0.014)		
$E_i \times (G_i = 1)$		−0.021**		−0.010	
		(0.010)		(0.016)	
$E_i \times (G_i = 2)$		−0.022**		−0.029	
		(0.009)		(0.018)	
$E_i \times (G_i = 3)$		−0.038***		−0.026	
		(0.010)		(0.020)	
$E_i \times (G_i = 4)$		−0.020**		−0.026	
		(0.008)		(0.019)	
$E_i \times (G_i = 5)$		−0.033***		−0.059**	
		(0.010)		(0.026)	
$Z_i \times (G_i = 1)$					−0.006
					(0.011)
$Z_i \times (G_i = 2)$					−0.016
					(0.010)
$Z_i \times (G_i = 3)$					−0.013
					(0.010)
$Z_i \times (G_i = 4)$					−0.011
					(0.009)
$Z_i \times (G_i = 5)$					−0.022**
					(0.010)

Note: E_i is the education indicator (staying in school to at least age 15), G_i is quintile of the educational attainment PGI. Columns (1) and (2) report OLS estimates, (3) to (5) IV estimates instrumenting E_i the binary instrument Z_i (born 1933 or later), and Column (5) the reduced form where frailty is regressed directly on interactions of Z_i and G_i. Columns (2) and (4) report the full effect of E_i within each PGI quintile as given by Equation 2. All specifications control for gender, survey wave, age, and mother's and father's age when leaving full-time education. IV specifications add the running variable (re-centered birth year) and its interaction with the instrument. Interacted specifications also include the first ten genetic principal components. Estimation sample as described in Section 3.2 with 18,705 observations from 3,415 individuals. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

frailty only for those in the highest PGI quintile. The coefficient increases in EA PGI and is about twice as large in quintile five.

4.3 Discussion

My main result, that the effect of education on old-age frailty increases in the EA PGI, mirrors findings for other outcomes. The effects of education on income and on cognitive ability in old age both increase in the EA PGI, driven by individuals at the top of the EA PGI distribution (Barcellos, Carvalho, and Turley 2021; Hollenbach, Schmitz, and Westphal 2026). I find that the effect of education on frailty rises by 1.2 percentage points per EA PGI quintile on average, compared to a large increase of 0.46 words per quintile for word recall (Hollenbach, Schmitz, and Westphal 2026). The effect on frailty, however, increases relatively smoothly in the EA PGI, while the effect on cognition evolves non-linearly. Studies of other health outcomes also find that the gains from education vary with genetic endowments, but they measure these endowments with PGIs for the specific condition rather than with the EA PGI. The

additional education from the 1972 UK schooling reform reduced Alzheimer's diagnoses, particularly for those at high genetic risk of the disease (Barcellos et al. 2025). It also improved body size, lung function, and blood pressure, mostly for those at the highest genetic risk of overweight (Barcellos, Carvalho, and Turley 2018). This study finds no differences by the EA PGI, but it measures health in middle age, whereas I study old age.

No study estimates the causal effect of education on old-age frailty with a comparable design. Huang et al. (2024) construct a Wald estimate from genetic discovery studies. For a set of genetic variants that predict schooling, they divide the effect of each variant on frailty (the "reduced form") by its effect on schooling (the "first stage") and obtain a LATE. Their estimate is large: one SD more schooling reduces frailty by about half an SD. My pooled estimate of the effect of schooling on frailty is roughly as large as the effect of retirement on the frailty index that Abeliansky and Strulik (2023) find for women. My estimate for the fifth EA PGI quintile (about 6 percentage points) exceeds their largest effect, a 4.5 percentage point reduction in frailty from retiring for men with low education.

More broadly, the evidence on the causal effect of education on later-life health in the UK is mixed. Clark and Royer (2013) estimate very small reduced-form effects of the 1947 and 1972 compulsory schooling reforms on self-rated health, and Jürges, Kruk, and Reinhold (2013) find no clear effects on self-rated health or blood-based markers of inflammation. In contrast, Silles (2009) reports sizable 2SLS estimates showing that education increases self-reports of good health by 6.5 percent, of having no long-term illness by 7.5 percent, and of having no activity-limiting illness by 5 percent. Using UK Biobank data and the 1972 reform, Davies et al. (2018) show that additional education reduced hypertension by 8 percent, diabetes by 3.6 percent, and strokes and heart attacks by about 1 percent each. It also lowered the 90th percentile of body size by 0.4 SD and the obesity rate by 8 percentage points, and raised the 30th percentile of blood pressure by a third of an SD (Barcellos, Carvalho, and Turley 2023). All of these studies observe individuals at younger ages than I do and focus on single health outcomes.

4.4 Distributional effects on frailty

An effect on the frailty index does not reveal where in the distribution it occurs. Where education reduces frailty matters for interpretation, because the frailest individuals in particular have lower quality of life, higher care needs, and lower chances of survival (Song, Mitnitski, and Rockwood 2010). I therefore split the frailty index into three commonly used categories (see, e.g., Song, Mitnitski, and Rockwood 2010; Davies et al. 2021): robust (frailty ≤ 0.08), pre-frail ($0.08 <$ frailty < 0.25), and frail (frailty ≥ 0.25), and use them as outcomes in my main IV model (Column 4 of Table 2). A lower probability of being frail would imply that individuals avoid the worst consequences of frailty, whereas a higher probability of being robust would mean more individuals thriving in old age.

Table 3 reports how the additional education shifts individuals between categories. By construction, a decrease in one category implies an increase in one or both of the others, and vice versa. Overall, the additional education decreases the probability of being frail. The decrease is especially large in the highest EA PGI quintile, where it is accompanied by sizable but statistically insignificant increases in

TABLE 3 – IV results at different margins of the frailty index

Dependent variable	Robust	Pre-frail	Frail
$E_i \times (G_i = 1)$	−0.014	0.072	−0.058
	(0.047)	(0.054)	(0.053)
$E_i \times (G_i = 2)$	−0.019	0.124**	−0.105*
	(0.055)	(0.063)	(0.059)
$E_i \times (G_i = 3)$	−0.016	0.134*	−0.118*
	(0.064)	(0.072)	(0.065)
$E_i \times (G_i = 4)$	0.060	0.009	−0.069
	(0.065)	(0.072)	(0.065)
$E_i \times (G_i = 5)$	0.103	0.079	−0.181**
	(0.091)	(0.097)	(0.083)
Observations	18,705	18,705	18,705
Individuals	3,415	3,415	3,415

Note: Dependent variables are indicators for being robust (frailty index ≤ 0.10), pre-frail (0.10 < frailty index ≤ 0.25), and frail (frailty index > 0.25). E_i is staying in school to at least age 15. IV estimates instrumenting E_i with exposure to the compulsory schooling reform (born 1933 or later). All specifications control for gender, survey wave, age, the running variable (re-centered birth year) and its interaction with the instrument. Estimation sample as described in Section 3.2. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

the probability of being robust (the lower end of the distribution) and of being pre-frail (the middle). In quintiles 2 and 3, the probability of being frail also decreases, but individuals move into the pre-frail rather than the robust category, which may partly explain the lack of a statistically significant effect on the overall index in these quintiles since for these individuals, the decrease was likely smaller than for those in the highest quintile.

The results suggest that the additional schooling reduced frailty at the upper end of its distribution, consistent with individuals avoiding or delaying the accumulation of a debilitating number of deficits. In the highest EA PGI quintile, it may also have kept individuals robust, that is, nearly deficit-free, for longer.

4.5 First-stage dynamics and selection on gains

Standard IV estimates of interaction effects do not necessarily reflect the true interaction effect. The estimator cannot separate changes in the treatment effect along the interaction variable (the interaction effect) from changes in the characteristics of compliers along the interaction variable (Hollenbach, Schmitz, and Westphal 2026). Hollenbach, Schmitz, and Westphal (2026) formalize two conditions that, if met, can cause 2SLS estimates to differ in size from the true interaction effect. First, compliers must differ in unobserved characteristics across strata of the interaction variable. In my setting, this would show up as differences in compliance behavior with the schooling reform by EA PGI. Second, treatment effects must exhibit essential heterogeneity, meaning that latent resistance to treatment is correlated with the individual treatment effect, so individuals select into treatment based

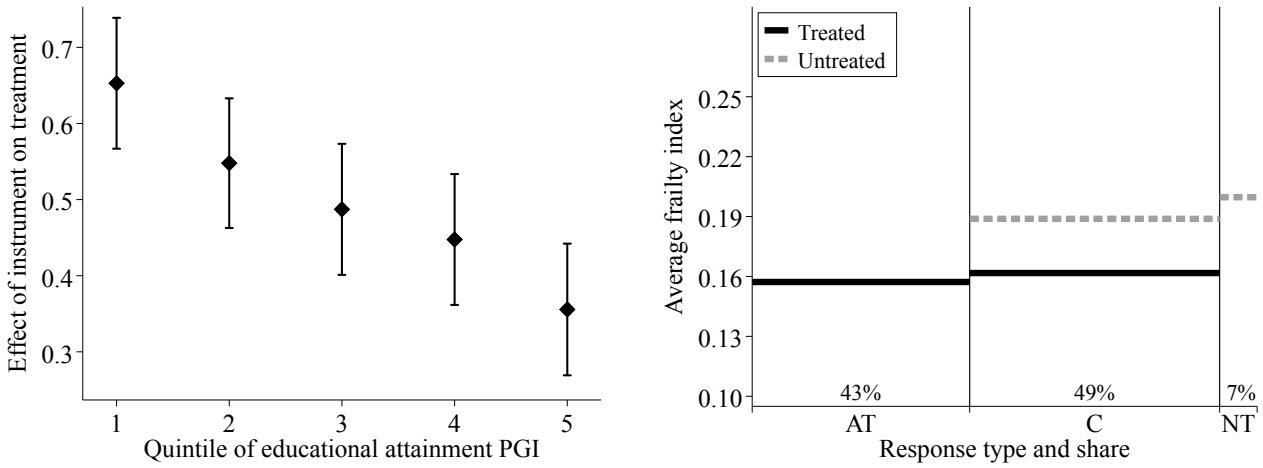
on unobserved gains (or losses) (see, e.g., Heckman, Urzua, and Vytlacil 2006). If both conditions apply, the 2SLS interaction estimate may compare potential outcomes of compliers at different points of latent resistance to treatment (condition 1). Since resistance is tied to differences in treatment effects (condition 2), the resulting LATE of the interaction may differ from the true LATE, which is obtained at a fixed point of latent resistance. The latter can be obtained by estimating the marginal treatment effect, which becomes necessary if the two conditions hold.

It is plausible that both conditions hold in my setting. First, the reaction to a schooling reform has been shown to differ by genetic endowment for educational attainment, e.g., in Barcellos, Carvalho, and Turley (2018) for the 1972 UK compulsory schooling reform and in Hollenbach, Schmitz, and Westphal (2026) for the 1947 reform I investigate. In the latter, those with high EA PGI are more likely to be always-takers (choose more education regardless of the reform) and those with low EA PGI are more likely to be compliers (choose more education because of the reform). Second, selection on gains has been established as a frequent feature of educational choices (Carneiro, Heckman, and Vytlacil 2011; Kamhöfer, Schmitz, and Westphal 2019; Westphal, Kamhöfer, and Schmitz 2022; Hollenbach, Schmitz, and Westphal 2026). Nevertheless, as Kowalski (2023) points out, even within the same (natural) experiment, patterns of selection into treatment can differ substantially depending on the outcome. In fact, Barcellos, Carvalho, and Turley (2021) do not find evidence of selection on gains in wage returns to the 1972 UK schooling reform.

Therefore, in the following, I investigate both conditions in my setting. First, Figure 5A shows the differences in the first-stage estimates by quintiles of the EA PGI. These correspond to the share of compliers in each quintile. There are sizable differences and a monotonic decline in the share of compliers along the EA PGI, from about 65 percent in the lowest PGI quintile down to 35 percent in the highest. The latter is still a sizable share, showing that the 1947 compulsory schooling reform had a broad impact. This entails that the first condition, characterized by differential first stages according to the interaction variable, applies.

Figure 5B presents visual evidence on the second condition, namely that treatment effects vary with unobserved resistance to taking more schooling. The figure shows the mean frailty index for groups that are informative about this latent resistance. Always-takers have the least resistance and take more schooling regardless of the reform. Never-takers have the most, as they do not take up the treatment even when affected by the reform. Compliers are in between. The figure does not stratify by PGI quintile because selection on gains, which I ultimately test for, is a pattern that affects, and can be shown in, the entire population.

The three groups are ordered on the horizontal axis by their latent resistance to schooling. The width of each group corresponds to its share of the sample. In my sample, 43 percent are always-takers, 49 percent are compliers, and 7 percent are never-takers. Treated outcomes are shown as solid black lines, untreated outcomes as dashed gray lines. The treated outcome of always-takers and the untreated outcome of never-takers can be obtained directly from the data, since I can identify these groups in the sample from their behavior. However, their counterfactual outcomes are never observed, since one group always takes the treatment and the other never does. For compliers, both treated and



A. First-stage differences by PGI quintile

B. Treated and untreated potential outcomes by type

FIGURE 5 – Evidence on differences in the first-stage and second-stage potential outcomes

Note: Panel A: Coefficients of Eq. 3 with 95% confidence intervals. They correspond to complier shares by PGI quintile. Instrument is born in or after 1933, treatment is left school at 15 or later. Appendix Table A.3 reports the exact estimates, standard errors, and first-stage F-statistics. Panel B: Treated outcome means for always-takers (AT) and compliers (C), and untreated means for compliers and never-takers (NT), without gene interactions. Black solid lines denotes treated, grey dashed lines untreated outcomes. The horizontal axis shows each group and its share of the sample, with the width of each group corresponding to that share. Groups are ordered from most to least willing to take the treatment (more education). Appendix Section B describes in detail how all numbers are calculated. Both analyses are calculated on the main sample described in Section 3.1.

untreated potential outcomes can be estimated, which I do using the method of Imbens and Rubin (1997). Appendix Section B describes in detail how all numbers in this section are computed.

Visual evidence of treatment effect heterogeneity by latent resistance to treatment would show up in Figure 5B as strong differences between the two treated outcome means, the two untreated outcome means, or both. The treated outcome means do not differ, but the untreated outcome means might. I therefore formally test both differences in means as well as selection on gains, following Brinch, Mogstad, and Wiswall (2017). Appendix Section B.1 describes this analysis in detail. Neither the treated nor the untreated potential outcomes differ significantly, and I find no statistically significant selection on gains. The untreated outcomes of compliers and never-takers differ by about one percentage point of frailty, which is not exactly zero. Consequently, the test of selection on gains, the difference in slopes through both pairs of potential outcomes, is not exactly zero either, but it is modest (-2.8 percentage points) and not statistically significant.

In education, always-takers typically have a larger treatment effect than compliers (Kamhöfer, Schmitz, and Westphal 2019; Hollenbach, Schmitz, and Westphal 2026). However, a comparison of treated potential outcomes alone (like in Figure 5B) cannot detect this, because treatment has already been chosen, so differences in treated outcomes also reflect selection (Kowalski 2023). Comparing LATEs, that is, differences between treated and untreated potential outcomes, remedies this, because the untreated potential outcome is the counterfactual unaffected by everything associated with treatment. Differences in LATEs between always-takers and compliers therefore reflect only treatment effect heterogeneity. Since the untreated potential outcome of always-takers is not observed, I calculate

a bound on it following Kowalski (2023). Appendix Section B.2 describes this analysis in detail. The bound implies a maximum possible LATE for always-takers. Comparing this maximum to the LATE of compliers shows that the two are practically the same size. This rules out that always-takers gain more from treatment than compliers.

A limitation of comparing treated outcomes is that differences between the treated potential outcomes of always-takers and compliers reflect both selection and treatment effect heterogeneity, because the untreated potential outcome of always-takers, which would provide a counterfactual unaffected by selection into treatment, is not observed (Kowalski 2023). To separate selection from heterogeneity, I calculate a bound on the untreated potential outcome of always-takers following Kowalski (2023). Appendix Section B.2 describes this analysis in detail. The bound also implies a maximum possible LATE for always-takers. Comparing this maximum to the LATE of compliers shows that the two are practically the same size (see Appendix Figure B.2). This rules out that always-takers gain more from treatment than compliers, a form of heterogeneity that is often a significant source of selection on gains in the education context.

Though the formal test for selection into gains does not yield a perfect zero, I interpret the results of both analyses as evidence that there is likely no selection on gains into schooling with respect to old-age frailty, especially compared to the strong evidence of selection on gains with respect to old-age cognition in Hollenbach, Schmitz, and Westphal (2026). Therefore, my 2SLS results should reflect the true interaction effects.

4.6 Robustness

I investigate the robustness of my main results with respect to the bandwidth, age composition, linearity of the cohort trends, and the choice of quintiles, the split for the EA PGI, account for attrition probability by reweighting, and investigate an alternative outcome related to functional aging without the mortality aspect. Table 4 reports the results.

Lowering the bandwidth to 8 and 6 years does not change the nature of the results. Only the coefficient for the highest quintile shows a statistically significant effect of a year of education on frailty. In Column (3), I lower the age threshold from 65 to 50. Because of the birth cohort restriction, this adds only 89 individuals. They still contribute many observations because the birth cohort restriction excludes younger respondents who joined ELSA later. Therefore, those who are added here entered early in the study and are observed over many waves. Adding these observations does not change the results.

Next, I use quartiles of the EA PGI instead of quintiles to estimate the interaction terms. The results show a coefficient close to zero for the lowest quartile of the EA PGI distribution and, again, a sizable reduction in frailty for the highest quartile. As expected, this effect is smaller than the one for the top 20 %. In this specification, there is also a statistically significant reduction in frailty of 0.04 from the additional year of education for the second quartile.

As discussed above in Section 3.4, attrition in old-age samples may interfere with treatment effects. To test for differential attrition, I first fill in missing waves for all individuals (after the first wave in which they are observed) up to wave 10, keeping only the observations in which the individual would have been within my age range of 65 to 85, so that aging out of the sample is not counted as attrition. I then create an attrition indicator equal to one if the individual did not respond in a given wave and use it as the outcome in two specifications: the pooled 2SLS estimation (without PGI interactions) and a reduced-form regression of attrition directly on the instrument and controls. Appendix Table A.4 reports the results. Both coefficients are close to zero and no statistically significant. So, the reform, and schooling induced by the reform, are unrelated to panel attrition in my sample.

Nevertheless, I also estimate the main regression with inverse probability weights that reweight for panel response (Table 4, Column 7). I obtain the individual-level weights from a logit regression of a response indicator (1 – *attrition*, as defined above) on observables, including the reform dummy, the education dummy, the EA PGI, birth cohort, a wave indicator, as well as age and frailty measured in the individual's first observed wave. The sample for this regression includes all waves after an individual's first appearance until wave 10, as long as their projected age still falls into the 65 to 85 window. Everyone's first wave is excluded, since the inclusion probability is one there. The estimate in Column (7) is my main regression, but weighted by the inverse of the predicted response probability from this logit regression. Again, the results are very similar to my main set of results.

Furthermore, I estimate a specification using only the first observation of each individual. By definition, there can be no panel attrition, since it only uses cross-sectional information. The results do not differ substantially from the main results.

Finally, I estimate the main regression for an alternative outcome, a sub-index of the frailty index that omits severe conditions. These omitted items are cardiovascular, metabolic, cognitive, and other severe diagnoses. The remaining 40 items include impairments in activities of daily living, self-assessments of physical and mental health, and cognitive tests, which cover functional and sensory aspects of the frailty index, but do not relate as much to the mortality dimension. The results are qualitatively and quantitatively (as a share of the standard deviation) comparable to the main results. This shows that the results hold for an alternative outcome that tracks mortality less closely, but instead measures quality of life and autonomy in old age.

4.7 Suggestive evidence on channels

Next, I turn to non-health outcomes that relate to potential channels of an effect of education on old-age health are affected by the reform. I investigate outcomes in three categories: income and pensions (Panel A of Table 5), employment history (Panel B), and health behavior (Panel C). The table also includes mean values of these outcomes and the number of observations and individuals. Each line is a separate IV regression of the respective outcome on my education indicator instrumented by the reform dummy without PGI interactions using the same sample limitations as the main analysis (10 birth cohorts to either side of the 1933 cutoff and observations from individuals aged 65–85).

TABLE 4 – Robustness

DV: Frailty index	(1) Main	(2) BW 8	(3) BW 6	(4) 50–85	(5) Quartiles	(6) IPW	(7) First obs.	(8) Functional
$E_i \times (G_i = 1)$	−0.010	−0.008	−0.017	−0.012	−0.009	−0.009	−0.007	−0.011
	(0.016)	(0.018)	(0.021)	(0.016)	(0.015)	(0.019)	(0.020)	(0.019)
$E_i \times (G_i = 2)$	−0.029	−0.027	−0.035	−0.030*	−0.041**	−0.030	−0.030	−0.027
	(0.018)	(0.020)	(0.022)	(0.018)	(0.018)	(0.021)	(0.023)	(0.022)
$E_i \times (G_i = 3)$	−0.026	−0.029	−0.029	−0.024	−0.019	−0.034	−0.032	−0.029
	(0.020)	(0.021)	(0.024)	(0.020)	(0.017)	(0.022)	(0.025)	(0.024)
$E_i \times (G_i = 4)$	−0.026	−0.028	−0.035	−0.025	−0.052**	−0.030	−0.018	−0.024
	(0.019)	(0.021)	(0.024)	(0.019)	(0.024)	(0.021)	(0.028)	(0.023)
$E_i \times (G_i = 5)$	−0.059**	−0.064**	−0.076**	−0.056**		−0.070**	−0.063*	−0.074**
	(0.026)	(0.030)	(0.035)	(0.026)		(0.029)	(0.033)	(0.032)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,705	16,057	12,962	21,083	18,705	18,705	3,415	18,705
Individuals	3,415	2,844	2,230	3,504	3,415	3,415	3,415	3,415

Note: The dependent variable is the frailty index. E_i is the education indicator (stayed in school until at least age 15) and G_i denotes quintiles of the educational attainment PGI. All columns report IV estimates of the interaction terms of the education indicator with each PGI group, instrumenting education with exposure to the 1947 compulsory schooling reform (born in 1933 or later). Column (1) reproduces the main IV estimates reported in Table 2. Columns (2) and (3) narrow the bandwidth around the reform cutoff to eight and six birth cohorts on either side. Column (4) widens the age range of the estimation sample to ages 50 to 85. Column (5) uses only the first observation per individual. Column (6) weights observations by the inverse of the estimated probability of being observed in a given wave, obtained from a logit model of panel response on baseline characteristics. Column (7) uses an alternative dependent variable, a frailty index constructed from the functional items only. Column (8) splits the PGI into quartiles instead of quintiles, so that only four interaction terms are estimated. All regressions include indicator variables for gender, survey wave, age, and mother's and father's age when they left full-time education, the running variable (re-centered birth year) and its interaction with the instrument, as well as the first ten principal components of the genetic data. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

The youngest respondents in my sample are 65 years old, so right at the official retirement age. Many have been retired since the first time they were observed in ELSA. Therefore, data on current jobs or earnings reflects jobs respondents have taken on in retirement, rather than the career these individuals have had. I deal with this issue in three ways. First, I use information from ELSA's life history module. There, respondents are asked about their careers and the earnings of their final job in their main career. Second, I look at subjective financial situation, that is, whether a person has ever experienced financial hardship. Third, I use information about the size of public and employer pensions as well as information about receiving a disability pension. The more a person has paid into public or private pension schemes, the more they should be able to take out, giving me a rough idea on whether income might have been affected.

Panel A suggests that the additional education from the 1947 schooling reform did not affect income or pensions. Coefficients on pay in final job, financial hardship, public and employer pensions are close to zero and not statistically significant. Overall, this seems consistent with the small to zero returns from this reform found by Devereux and Hart (2010) or Clark (2023). However, there is a sizable decrease in the probability of receiving a disability pension. Abeliansky and Strulik (2023) show an occupational gradient in the frailty index. Workers in physically more demanding jobs accumulate health deficits faster. A decrease in the probability of relying on disability pension may hint at less burden and injuries through the job. Unfortunately, ELSA does not provide detailed information about

TABLE 5 – Effect of education on non-health outcomes

	IV estimate	Mean	Observations	Individuals
A. Income and pensions				
Log final pay in last job	−0.117	16,759	2,118	2,118
	(0.236)			
Ever severe financial hardship	−0.024	0.177	3,133	3,133
	(0.055)			
Log public old-age pension income	0.042	5,704	30,941	6,711
	(0.150)			
Log private/employer pension income	0.749	4,343	30,940	6,697
	(0.719)			
Receives disability pension	−0.066**	0.137	31,251	6,716
	(0.032)			
B. Employment history				
Ever unemployed	−0.014	0.342	3,603	3,603
	(0.064)			
Ever self-employed	0.015	0.206	3,579	3,579
	(0.054)			
Ever part-time	0.023	0.475	3,579	3,579
	(0.057)			
Number of jobs	−0.008	5.294	3,603	3,603
	(0.343)			
Early retirement	0.016	0.533	5,927	5,927
	(0.056)			
C. Health behavior				
Ever smoked	−0.138**	0.606	3,601	3,601
	(0.064)			
Smokes currently	−0.047	0.098	31,285	6,678
	(0.031)			
Drank alcohol last 12 months	−0.038	0.845	27,205	6,487
	(0.038)			
Risky drinking (15+/week)	−0.013	0.056	31,527	6,730
	(0.018)			
Any light activity	0.027	0.874	31,527	6,730
	(0.026)			
Any moderate activity	0.126***	0.755	31,527	6,730
	(0.036)			
Any vigorous activity	0.054	0.304	31,527	6,730
	(0.036)			
BMI	−0.538	27.976	11,426	4,951
	(0.616)			

Note: Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

the nature of the jobs in the life history module, like occupational classifiers. Therefore, I am unable to investigate patterns of physical or mental strain of the former jobs.

In terms of employment history, I am able to identify instances of unemployment, self-employment, part-time emplyment, and the number of jobs a person has held. I also investigate the impact of education on taking early retirement, identified from the main sample. The results suggest that

there is no effect of the additional year of compulsory education on whether a respondent was ever unemployed, self-employed, ever in part time or on the number of jobs they have had during their working life or has retired early. The literature broadly finds the same: Clark (2023) investigates the effect of the 1947 compulsory schooling reform on several labor market outcomes, including labor force participation, unemployment, selfemployment and finds no effects.

Panel C investigates different aspects of (risky) health behaviors. I observe a large decrease in the probability of ever having smoked and a statistically insignificant, but still sizable estimate suggesting a reduction in the probability of currently smoking. However, there does not seem to be an effect on current alcohol consumption or heavy (risky) drinking, which, according to official UK advisory, is more than 14 units per week (UK Department of Health and Social Care 2016). In terms of physical activity, there is no effect of education on doing any light activity, but sizable positive effects on doing moderate and vigorous activity. Nevertheless, this does not seem to translate to weight reductions, as the estimate on BMI is not statistically significant and not of relevant size given the high average.

Better health behavior may be able to explain reductions in old-age frailty from increased education. Strulik (2018a) uses a health deficit model centered around frailty and calibrated to US data to show that more educated inviduals delay the accumulation of deficits through lower unhealthy consumption (e.g. smoking) and increased health investments (e.g. exercise). In Strulik (2018b), he shows that smoking drives long-run health trajectories and finds longevity gains from not smoking. Lack of exercise has long been identified as a key risk factor for old-age morbidity (World Health Organization 2009). Futhermore, evidence from RCTs randomizing incentives for exercise and exercise itself show that increased physical activity reduces blood sugar in middle-aged diabetics and helps manage the disease (Aggarwal, Dizon-Ross, and Zucker 2026) and increases functional health in old age (Pahor et al. 2014). Furthermore, both increased exercise and reduced smoking are frequently reported as key mechanisms for the effect of retirement on old-age health (see, e.g. Insler 2014 or Eibich 2015).

4.8 Conclusion

In this study, I examine how the effect of education on physical health in old age varies with genetic propensity for educational attainment, using data on individuals aged 65–85 from the English Longitudinal Study of Ageing. First, I document a sizable genetic gap in old-age frailty. The difference in frailty between the lowest and the highest quintile of the EA PGI is about as large as the difference between the lowest and the fourth quintile of socioeconomic status. Second, I estimate the causal effect of education on old-age frailty by EA PGI in a fuzzy regression discontinuity design, using the 1947 raising of the UK school leaving age from 14 to 15 as exogenous variation in schooling. The health returns to schooling rise monotonically along the EA PGI, from zero in the lowest quintile to a large 6 percentage point reduction in frailty in the highest.

This shows that an early-life interaction between an exogenous increase in schooling and genetic propensity contributes to larger health differences late in life. Such a complementarity between genes and educational attainment matches the notion that those with a higher genetic predisposition for education extract larger benefits from their education (Houmark, Ronda, and Rosholm 2024). It is

also in line with findings that the effect of education on income and on cognitive ability in old age increase in the EA PGI (Barcellos, Carvalho, and Turley 2021; Hollenbach, Schmitz, and Westphal 2026). On the other hand, increasing schooling seems to reduce the impact of genetic *risk* for specific conditions, including overweight and Alzheimer's disease (Barcellos, Carvalho, and Turley 2018; Barcellos et al. 2025). These findings do not conflict. The first refers to skill formation, where only those at the upper end of genetic propensity for schooling can translate additional instruction time later-life outcomes. The second refers to compensating effects of additional education, which could, for example, materialize through better health knowledge.

This study shows that a large-scale schooling policy increased the genetic health gradient by reduced frailty mainly for the top of the EA PGI distribution. Nevertheless, it reduced genetic differences in educational attainment by moving more individuals from the lower part of the EA PGI distribution into additional schooling (see the first-stage results in Figure 5A and Hollenbach, Schmitz, and Westphal 2026, who cover this in more detail). Yet, this larger increase in additional schooling did not lead to imrovements in later-life health. This does not imply that additional schooling at other margins, such as tertiary education, could not benefit low-PGI students disproportionately.

I do not interpret my results as evidence that the EA PGI can help target educational policies toward individuals at the lower end or a group that would be able to most effectively utilize the change. Apart from ethical concerns about collection of genetic data from every child, most PGIs are currently constructed from samples of European ancestry and their validity in broader populations is unclear (Martin et al. 2019). Moreover, their predictive power at the individual level is limited (Okbay et al. 2022). Nevertheless, they are a useful, albeit imperfect, proxy for the ability to learn and can show where policies fail or where they are particularly effective in changing behavior and outcomes.

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Appendix A. Additional figures and tables

TABLE A.1 – Frailty index items and coding

Description	Coding
Difficulty with walking 100 yards	No=0, Yes=1
Difficulty sitting for about two hours	No=0, Yes=1
Difficulty getting up from a chair after sitting for long periods	No=0, Yes=1
Difficulty climbing several flights of stairs without resting	No=0, Yes=1
Difficulty climbing one flight of stairs without resting	No=0, Yes=1
Difficulty stooping, kneeling, or crouching	No=0, Yes=1
Difficulty reaching or extending arms above shoulder level	No=0, Yes=1
Difficulty pulling or pushing large objects like a living room chair	No=0, Yes=1
Difficulty lifting or carrying weights over 10 pounds, like a heavy bag	No=0, Yes=1
Difficulty picking up a 5p coin from a table	No=0, Yes=1
Difficulty dressing, including putting on shoes and socks	No=0, Yes=1
Difficulty walking across a room	No=0, Yes=1
Difficulty bathing or showering	No=0, Yes=1
Difficulty eating, such as cutting up your food	No=0, Yes=1
Difficulty getting in or out of bed	No=0, Yes=1
Difficulty using the toilet, including getting up or down	No=0, Yes=1
Difficulty using a map to figure out how to get around in a strange place	No=0, Yes=1
Difficulty preparing a hot meal	No=0, Yes=1
Difficulty shopping for groceries	No=0, Yes=1
Difficulty making telephone calls	No=0, Yes=1
Difficulty taking medications	No=0, Yes=1
Difficulty managing money (e.g. paying bills, keeping track of expenses)	No=0, Yes=1
Difficulty doing work around the house or garden	No=0, Yes=1
Self-reported general health	Excellent=0, v.good=0.25, Good=0.5, Fair=0.75, Poor=1
Felt depressed much of the time during the past week	No=0, Yes=1
Felt everything done during the past week was an effort	No=0, Yes=1
Felt sleep was restless much of the time during the past week	No=0, Yes=1
Was happy much of the time during the past week	Yes=0, No=1
Felt lonely much of the time during the past week	No=0, Yes=1
Enjoyed life much of the time during the past week	Yes=0, No=1
Felt sad much of the time during the past week	No=0, Yes=1
Could not get going much of the time during the past week	No=0, Yes=1
High blood pressure or hypertension	No=0, Yes=1
Angina	No=0, Yes=1
Heart attack (including MI or coronary thrombosis)	No=0, Yes=1
Congestive heart failure	No=0, Yes=1
Abnormal heart rhythm	No=0, Yes=1
Diabetes or high blood sugar	No=0, Yes=1
Stroke (cerebral vascular disease)	No=0, Yes=1
Chronic lung disease (e.g. chronic bronchitis or emphysema)	No=0, Yes=1
Asthma	No=0, Yes=1
Arthritis (including osteoarthritis or rheumatism)	No=0, Yes=1
Osteoporosis	No=0, Yes=1
Cancer or malignant tumor (excluding minor skin cancers)	No=0, Yes=1

Continued on next page

Table A.1 continued from previous page

Description	Coding
Parkinson's disease	No=0, Yes=1
Any emotional, nervous or psychiatric problems	No=0, Yes=1
Alzheimer's disease	No=0, Yes=1
Dementia, organic brain syndrome, senility or other serious memory impairment	No=0, Yes=1
Self-reported eyesight function (with lenses, if appropriate)	Excellent=0, v.good=0.2, Good=0.4, Fair=0.6, Poor=0.8, Blind=1
Self-reported hearing function (with hearing aid, if appropriate)	Excellent=0, v.good=0.25, Good=0.5, Fair=0.75, Poor=1
Identify today's date: day of month	No=0, Yes=1
Identify today's date: month	No=0, Yes=1
Identify today's date: year	No=0, Yes=1
Identify the day of the week	No=0, Yes=1
Immediate word recall (quartiles)	Q1=1, Q2=0.67, Q3=0.33, Q4=0
Delayed word recall (quintiles)	Q1=1, Q2=0.75, Q3=0.5, Q4=0.25, Q5=0

TABLE A.2 – Summary statistics (full table)

	Full sample				By E_i (Left school at 15 or later)		
	Mean	SD	Min	Max	0	1	Diff. (p-val.)
Frailty index	0.162	0.110	0.000	0.772	0.194	0.153	−0.041 (0.000)***
Left school at 15 or later	0.790		0	1	0.000	1.000	1.000
Born 1933 or later	0.685		0	1	0.143	0.829	0.686 (0.000)***
Educational attainment PGI							
1st quintile	0.186				0.234	0.174	−0.061 (0.000)***
2nd quintile	0.199				0.214	0.195	−0.019 (0.008)***
3rd quintile	0.200				0.189	0.203	0.014 (0.048)**
4th quintile	0.209				0.206	0.210	0.004 (0.582)
5th quintile	0.206				0.158	0.219	0.061 (0.000)***
Female	0.544		0	1	0.525	0.549	0.024 (0.009)***
Age	74.238	5.344	65.000	85.000	77.049	73.491	−3.559 (0.000)***
Birth year	1,935.027	5.241	1,923.000	1,943.000	1,929.798	1,936.416	6.619 (0.000)***
Age mother left education							
≤ 14	0.678				0.656	0.683	0.027 (0.001)***
15	0.060				0.018	0.071	0.053 (0.000)***
16	0.039				0.007	0.048	0.040 (0.000)***
17	0.013				0.002	0.016	0.015 (0.000)***
18	0.017				0.004	0.020	0.016 (0.000)***
≥ 19	0.019				0.005	0.022	0.017 (0.000)***
Missing	0.175				0.308	0.140	−0.168 (0.000)***
Age father left education							
≤ 14	0.663				0.634	0.671	0.037 (0.000)***
15	0.054				0.022	0.062	0.039 (0.000)***
16	0.044				0.005	0.054	0.049 (0.000)***
17	0.010				0.004	0.011	0.008 (0.000)***
18	0.019				0.005	0.022	0.018 (0.000)***
≥ 19	0.030				0.005	0.037	0.031 (0.000)***
Missing	0.181				0.325	0.143	−0.182 (0.000)***
Survey wave							
1	0.104				0.188	0.081	−0.107 (0.000)***
2	0.125				0.189	0.108	−0.082 (0.000)***
3	0.124				0.164	0.114	−0.051 (0.000)***
4	0.146				0.152	0.145	−0.008 (0.226)
5	0.131				0.118	0.135	0.017 (0.006)***
6	0.114				0.087	0.122	0.035 (0.000)***
7	0.093				0.055	0.103	0.048 (0.000)***
8	0.075				0.030	0.087	0.057 (0.000)***
9	0.058				0.012	0.071	0.059 (0.000)***
10	0.029				0.004	0.036	0.032 (0.000)***
PC1	−0.000	0.012	−0.027	0.041	−0.001	−0.000	0.000 (0.120)
PC2	0.000	0.012	−0.027	0.030	−0.000	0.000	0.000 (0.039)**
PC3	0.000	0.011	−0.057	0.047	0.000	0.000	0.000 (0.261)
PC4	0.000	0.012	−0.036	0.054	0.000	0.000	−0.000 (0.543)
PC5	0.000	0.012	−0.039	0.040	0.000	0.000	−0.000 (0.088)*
PC6	0.000	0.012	−0.034	0.044	−0.001	0.000	0.001 (0.000)***
PC7	0.000	0.012	−0.034	0.042	−0.001	0.000	0.001 (0.000)***
PC8	0.000	0.012	−0.039	0.042	0.000	−0.000	−0.000 (0.186)
PC9	0.000	0.012	−0.039	0.040	−0.000	0.000	0.000 (0.068)*
PC10	−0.000	0.012	−0.048	0.048	−0.000	−0.000	0.000 (0.603)
Observations			18,705		3,927	14,778	

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

TABLE A.3 – Reduced-form and first-stage estimates

	DV: Left school at 15 or later (E_i)	F
$Z_i \times (G_i = 1)$	0.653***	221.6
	(0.044)	
$Z_i \times (G_i = 2)$	0.548***	159.1
	(0.043)	
$Z_i \times (G_i = 3)$	0.487***	123.1
	(0.044)	
$Z_i \times (G_i = 4)$	0.448***	104.2
	(0.044)	
$Z_i \times (G_i = 5)$	0.356***	64.9
	(0.044)	
Observations	18,705	
Individuals	3,415	

Note: First stages estimated according to Eq. 3 on the main sample described in Section 3.1. Each interaction of the instrument Z_i with a PGI quintile G_i yields a separate first stage (though estimated in one model). The second column reports F-statistics for the excluded instrument, separately for each interaction. Controls: linear cohort trend, its interaction with the instrument, indicator variables for gender, survey wave, age, and mother's and father's age when they left full-time education, and the first ten principal components of the genetic data. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

TABLE A.4 – Panel attrition

DV: Attrition	(1) 2SLS	(2) RF
E_i	−0.007	
	(0.037)	
Z_i		−0.003
		(0.020)
Controls	Yes	Yes
Observations	25,805	25,805
Individuals	3,415	3,415

Note: Notes: The dependent variable is an indicator equal to one if an individual eligible for interview in a given wave does not respond. E_i is the education indicator (stayed in school until at least age 15), instrumented by Z_i in column (1). Z_i is an indicator for exposure to the 1947 compulsory schooling reform (born in 1933 or later). Column (1) reports the 2SLS estimate, column (2) the reduced-form estimate. All specifications control for the running variable, its interaction with Z_i, gender, and wave fixed effects. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

Appendix B. Additional analyses and technical details

B.1 Testing for selection on gains

This section describes how I calculate each of the numbers reported in Figure 5B, namely the shares of always-takers (AT), compliers (C), and never-takers (NT) and their four potential outcome means. Furthermore, I run a formal test of the slopes in potential outcomes that make up the test for selection on gains based on Brinch, Mogstad, and Wiswall (2017).

E_i denotes the binary treatment (in school until at least age 15), Z_i the binary instrument (born 1933 or later, i.e., exposed to the reform), c_i the running variable (birth year, centered at the reform cutoff), and X_i the vector of common controls from my main specification (see Section 3.3). All quantities are identified under the standard LATE assumptions, in particular instrument exogeneity and monotonicity (no defiers), and follow Imbens and Rubin (1997). I focus on how to calculate these quantities here. Appendix G of Hollenbach, Schmitz, and Westphal (2026), for example, includes a more detailed step-by-step derivation.

Essentially, two regressions provide everything I need. For the type shares, I estimate the following first stage

$$E_i = \zeta_0 + \zeta_1 Z_i + \gamma\, c_i + X_i'\eta + \kappa_i, \tag{4}$$

This corresponds to Eq. 3 but without interactions with the EA PGI. For the outcome means, I regress the frailty index on a full set of interactions between E_i and Z_i without an intercept:

$$\begin{aligned} Y_i = {} & \lambda_1 \mathbb{1}\{E_i = 1\}\,\mathbb{1}\{Z_i = 0\} \\ & + \lambda_2 \mathbb{1}\{E_i = 1\}\,\mathbb{1}\{Z_i = 1\} \\ & + \lambda_3 \mathbb{1}\{E_i = 0\}\,\mathbb{1}\{Z_i = 0\} \\ & + \lambda_4 \mathbb{1}\{E_i = 0\}\,\mathbb{1}\{Z_i = 1\} \;+\; \delta\, c_i + X_i'\mu + \nu_i. \end{aligned} \tag{5}$$

Both regressions are evaluated at the cutoff, $c_i = 0$, and adjusted for the common controls X_i at their sample means. In what follows, I distinguish observable quantities latent ones from compliers, who cannot be identified at the individual level.

Observed types. The following six quantities can be directly obtained from the data, via the two regression I run:

i. **Share of always-takers, ϕ_{AT}.** Individuals who take the treatment although the instrument is switched off must, under monotonicity, be always-takers, so $\phi_{AT} = \Pr(E_i = 1 | Z_i = 0)$. This is the intercept ζ_0 in Eq. 4.
ii. **Share of never-takers, ϕ_{NT}.** Analogously, those who do not take the treatment although the instrument is switched on must be never-takers, so $\phi_{NT} = \Pr(E_i = 0 | Z_i = 1) = 1 - \zeta_0 - \zeta_1$.

iii. **Treated potential outcome for always-takers,** $\mathbb{E}[Y_i^1|AT]$**.** Since the group with $E_i = 1, Z_i = 0$ consists only of always-takers and under instrument exogeneity, their mean frailty identifies this potential outcome via the coefficient λ_1 in Eq. 5.

iv. **Untreated potential outcome for never-takers,** $\mathbb{E}[Y_i^0|NT]$**.** Likewise, the group with $E_i = 0, Z_i = 1$ consists only of never-takers, so their mean frailty, the coefficient λ_4, identifies this potential outcome.

v. **Treated mean of always-takers and compliers.** The coefficient λ_2 is the mean frailty of those who take the treatment when the instrument is switched on. This group mixes always-takers and compliers, who cannot be distinguished at the individual level. The mean needed as an input to calculate the treated complier mean below.

vi. **Untreated mean of never-takers and compliers.** Analogously, the coefficient λ_3 is the mean frailty of those who do not take the treatment when the instrument is switched off, a mix of never-takers and compliers. It is the input needed for the untreated complier mean.

Latent type. Compliers are latent, i.e., cannot be identified in the data directly. When they are observed, they are pooled with always-takers (if $Z_i = 1$) or with never-takers (if $Z_i = 0$). Their share and their two potential outcomes are therefore obtained residually:

vii. **Share of compliers,** ϕ_C**.** Under monotonicity there are only three types, and their shares must add to one, so $\phi_C = 1 - \phi_{AT} - \phi_{NT}$. This quantity is identified by first-stage coefficient ζ_1 in Eq. 4.

viii. **Treated potential outcome for compliers,** $\mathbb{E}[Y_i^1|C]$**.** The mixed treated mean (v.) is a weighted average of the treated means of always-takers and compliers. Solving for the complier part shows the treated potential outcome can be calculated only based on already established quantities:

$$\mathbb{E}[Y_i^1|C] \;=\; \lambda_2 \frac{\phi_{AT} + \phi_C}{\phi_C} \;-\; \mathbb{E}[Y_i^1|AT] \frac{\phi_{AT}}{\phi_C}, \tag{6}$$

ix. **Untreated potential outcome for compliers,** $\mathbb{E}[Y_i^0|C]$**.** The same logic applies to the mixed untreated mean (vi.). Again solving for the complier part yields:

$$\mathbb{E}[Y_i^0|C] \;=\; \lambda_3 \frac{\phi_{NT} + \phi_C}{\phi_C} \;-\; \mathbb{E}[Y_i^0|NT] \frac{\phi_{NT}}{\phi_C}. \tag{7}$$

The difference between (6) and (7) is the local average treatment effect.

Slopes in potential outcomes and selection on gains. The three types can be ordered according to their unobserved resistance to treatment U^E. Represented on the unit interval, always-takers lie furthest to the left (have the lowest resistance to treatment) between 0 and ϕ_{AT}, compliers in the middle between ϕ_{AT} and $\phi_{AT} + \phi_C$, and never-takers on the right (highest resistance to treatment) between $\phi_{AT} + \phi_C$ and 1. A plain difference between two of these means ignores the group sizes and thus how far they are apart on the unit interval, which matters for calculating a slope. The same difference in frailty is a much more drastic (therefore steeper) change when it exists between two narrow groups than when it stretches across two large ones. Following Brinch, Mogstad, and Wiswall (2017), I treat each

type-specific mean as if it were measured in the middle of that type's interval. The horizontal distance between two means is therefore half the sum of the two type shares. Dividing the difference in means by that distance gives the slope. Since the binary instrument yields only two means per potential outcome, always-takers and compliers when treated, compliers and never-takers when untreated, I additionally have to assume that the potential outcomes are linear in U^E. I calculate:

i. **Slope of the treated potential outcome,** $\partial\,\mathbb{E}[Y_i^1|U^E = u]/\partial u$**.** From the two treated means, always-takers and compliers:
$$\frac{\mathbb{E}[Y_i^1|C] - \mathbb{E}[Y_i^1|AT]}{(\phi_{AT} + \phi_C)/2}.$$

ii. **Slope of the untreated potential outcome,** $\partial\,\mathbb{E}[Y_i^0|U^E = u]/\partial u$**.** From the two untreated means, compliers and never-takers:
$$\frac{\mathbb{E}[Y_i^0|NT] - \mathbb{E}[Y_i^0|C]}{(\phi_C + \phi_{NT})/2}.$$

iii. **Slope of the treatment effect,** $\partial\,\mathbb{E}[Y_i^1 - Y_i^0|U^E = u]/\partial u$**.** This is the difference of the two slopes above, reported in the last column of Table B.1. The treatment effect at any point of unobserved resistance to the treatment u is the gap between the treated and the untreated potential outcome, so the difference of their slopes says whether this gap widens or narrows as resistance to treatment increases. No slope means the gap is constant and treatment effects do not depend on the unobserved resistance to treatment, and there is no essential heterogeneity. A negative value means treatment effects shrink as resistance rises, which implies selection on gains. A positive slope would mean selection on losses.

Table B.1 shows the means and shares from Figure 5B, as well as tests for the difference between treated and untreated means, for slopes in treated and untreated outcomes (which additionally account for cohort size), and for selection on gains. The latter tests are based on Brinch, Mogstad, and Wiswall (2017). The difference between each pair of outcomes is small and not statistically significant. So are the slopes in potential outcomes as well as their difference, which yields a test for selection on gains across all groups. A negative coefficient on this test implies selection on gains (whereas a positive would imply selection on losses), meaning that those at the left of the horizontal axis in Figure 5B, the always-takers with the strongest taste for education, also benefit more from it (i.e. have higher treatment effects), and returns fall as willingness to take more education declines, that is, to the right. However, as Table B.1 shows, the estimate for selection on gains is close to zero and not statistically significant.

Standard errors. I obtain standard errors by resampling rather than analytically. I use the Bayesian bootstrap of Rubin (1981) with 1,000 replications. In each replication, every individual receives a weight drawn from an exponential distribution with mean one and normalized to sum to one. The weight is constant within an individual. The full procedure, both regressions, the type shares, the complier means, and the slopes, is re-estimated on each set of weights, and the reported standard error is the

TABLE B.1 – Frailty by response type: potential outcomes and selection on gains

Type	Share	Potential outcome	Mean frailty	Difference of means	Slope in potential outcomes	Selection on gains
Always-taker	0.431	Treated AT	0.157	0.004	0.010	
		Treated C	0.162	(0.015)	(0.031)	
Complier	0.495					-0.028
		Untreated C	0.189	0.011	0.038	(0.051)
Never-taker	0.074	Untreated NT	0.200	(0.015)	(0.053)	

Note: Shares of always-takers (AT), compliers (C), and never-takers (NT) alongside treated potential outcome means for always-takers and compliers, untreated potential outcome means for compliers and never-takers, as well as treated and untreated potential-outcome tests and a test for selection on gains. All estimates use the main sample described in Section 3.1. The shares come from a first-stage regression of the treatment (in school until at least 15) on the instrument (born 1933 or later), without gene interactions. The means are based on a regression of the frailty index on interactions of the treatment with the instrument. Appendix Section B includes a detailed description of how each number is calculated. Standard errors are computed using the Bayesian bootstrap (Rubin 1981) with 1,000 replications and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

standard deviation of the resulting distribution of estimates. The Bayesian bootstrap is smooth in the weights, which is convenient here because the quantities of interest involve ratios of estimated shares.

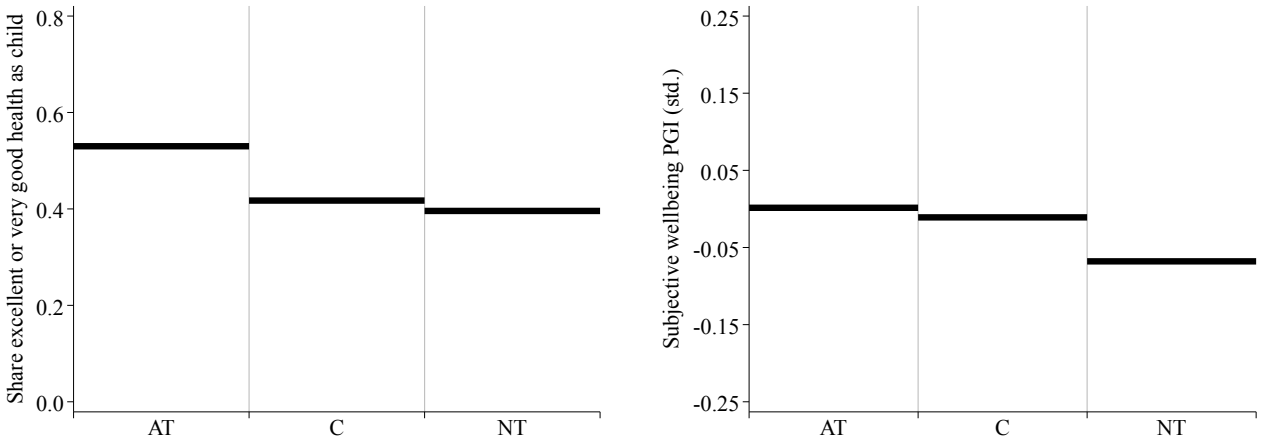
B.2 Bounding a LATE for always-takers

Comparing treated outcomes of always-takers and compliers involves comparing the average frailty of those most eager to take on more education and do so regardless of the reform to the average of those who were prompted by the reform to take an additional year of education. This comparison comprises both differences in selection and treatment effect heterogeneity (Kowalski 2023). Panel B of Figure 5 shows no difference in these treated outcomes. But since we cannot separate the two channels, this could indicate either no selection and no heterogeneity, or that both work in opposite directions.

Therefore, I make an additional assumption in the spirit of Kowalski (2023). I assume weak monotonicity in untreated outcomes related to willingness to take the treatment. This means that, in the absence of the treatment, the frailty of always-takers is at least as low as that of compliers, whose frailty is at least as low as that of never-takers. This assumption should be unproblematic, especially when compared to assuming the same for treated outcomes, where effect heterogeneity would complicate the ordering. At worst, it implies that – absent the treatment – there should be no differences in frailty between always-takers, compliers and never-takers.

Comparing pre-treatment covariates related to the outcome across the three groups can be used to further support the credibility of this ordering (Kowalski 2023). In my case, this is not easy to achieve, since treatment occurred when ELSA participants were young. In Appendix Figure B.1 I plot shares of respondents recalling being in excellent or very good health as a child. A larger share of AT report having been very healthy than C and NT, confirming the ordering. However, this measure was asked retrospectively and refers only to "childhood", and thus may also reflect treated periods. Therefore, I also include means of the standardized subjective wellbeing PGI. This is set at conception, but does not refer to realized outcomes due to its probabilistic nature. Nevertheless, a higher PGI predicts better

subjective wellbeing. There are few differences between the groups, but in general, the order fits, with AT having at least as many variants predicting wellbeing as C, who have slightly more variants than NT.



A. Shares of AT, C, and NT who recall very good childhood health

B. Means of wellbeing PGI for AT, C, and NT

FIGURE B.1 – Support for weak monotonicity in untreated outcomes

Note: Both analyses are calculated on the main sample described in Section 3.1.

What the assumption of weak monotonicity in untreated outcomes allows for is the use of the untreated complier mean as an upper bound for the unobserved untreated potential outcome for always-takers, which also implies an upper bound on the local average treatment effect for always-takers. This is informative about treatment effect heterogeneity because I can now compare two LATEs. Figure B.2 adds the bound to Figure 5B, yielding a visual representation of both LATEs. This bounding exercise shows that the maximum possible LATE for always-takers has the same size as the point-identified LATE for compliers. However, if there were selection on gains, we would expect always-takers (who are more eager to take the treatment) to have a larger LATE.

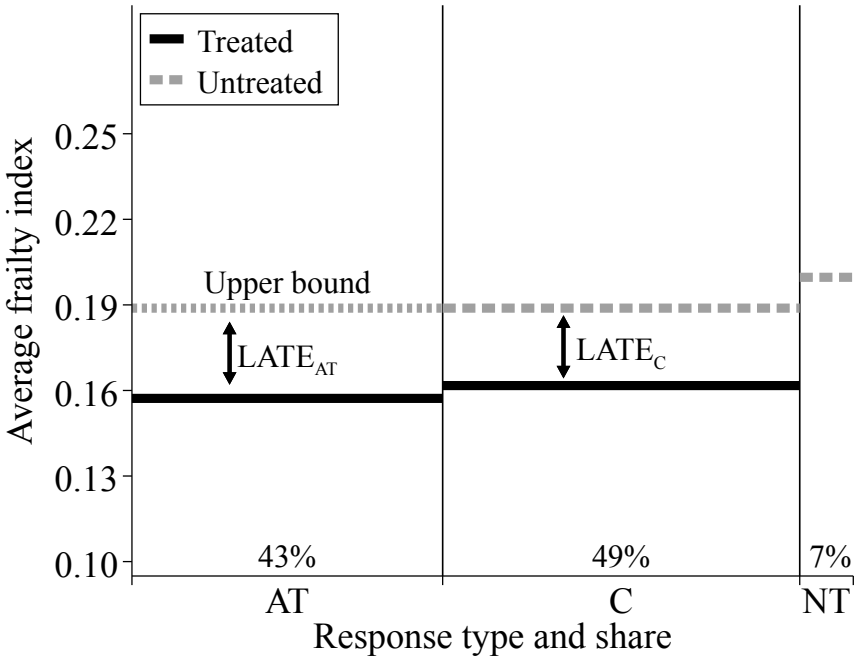


FIGURE B.2 – Treated and untreated potential outcomes by type

Note: Treated outcome means for always-takers (AT) and compliers (C), and untreated means for compliers and never-takers (NT), without gene interactions. Black solid lines denotes treated, grey dashed lines untreated outcomes. The horizontal axis shows each group and its share of the sample, with the width of each group corresponding to that share. Groups are ordered from most to least willing to take the treatment (more education). Additionally assuming weak monotonicity of untreated potential outcomes yields an upper bound on the local average treatment effect for always-takers. Calculated on the main sample described in Section 3.1.